

REFLECTION OF SOUND BY A BOUNDARY LAYER

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A method is given for the calculation of the reflection coefficient for plane waves incident upon a plane boundary when the acoustic medium flows parallel to the plane and the resulting boundary layer is of finite thickness. Results are given for limited ranges of the important parameters, which are the Mach number of the flow, the angle of incidence, the thickness of the boundary layer, and the acoustic admittance of the solid boundary.

The propagation of sound is profoundly affected by the gross motion of the acoustic medium. If this motion is a uniform steady flow, the analysis can often be carried out by considering a moving co-ordinate system [1], but if the flow is non-uniform, no simplification of the governing equations is achieved in this way. If the flow streamlines are all parallel (in the x -direction, say), the acoustic pressure must satisfy

$$\left(\frac{1}{c} \frac{\partial}{\partial t} + M \frac{\partial}{\partial x}\right) \nabla^2 p - \left(\frac{1}{c} \frac{\partial}{\partial t} + M \frac{\partial}{\partial x}\right)^3 p - 2 \frac{\partial M}{\partial y} \frac{\partial^2 p}{\partial x \partial y} - 2 \frac{\partial M}{\partial z} \frac{\partial^2 p}{\partial x \partial z} = 0. \quad (1)$$

In this equation, $M(y, z)$ is the Mach number of the steady flow, c is the speed of sound, assumed to be constant, x, y, z, t are the independent variables of space and time, and p is the acoustic pressure.

A number of investigators, for example, Swinbanks [2], Mungur and Plumblee [3], and Eversman [4], have used equation (1), or its equivalent in cylindrical co-ordinates, as a starting point for the calculation of sound propagation in ducts with non-uniform flow. The simpler case of the sound field in a half-space bounded by a single plane, with a flow parallel to that plane, seems to have been overlooked. This, of course, is an idealization of sound propagation in the open atmosphere with the wind blowing over level ground. Lord Rayleigh discussed this problem briefly [5], showing how the path of sound is refracted by the non-uniform wind.

In this paper, it is assumed that the flow is uniform, except in a boundary layer of constant thickness, δ . Throughout the uniform flow the Mach number is M_0 . In the boundary layer, the Mach number varies continuously from zero at the wall, to M_0 at the distance δ . The interface between the boundary layer and the uniform flow is taken to be the co-ordinate plane $y = 0$; thus the wall is $y = -\delta$. Since M is a function of y only, the last term in equation (1) vanishes.

For this situation

$$p = P(y) \exp [i(\omega t - k_x x - k_z z)] \quad (2)$$

is a solution of equation (1), provided $P(y)$ satisfies

$$(K - k_x M)P'' + 2k_x M'P' + (K - k_x M)[(K - k_x M)^2 - k_x^2 - k_z^2]P = 0. \quad (3)$$

In equation (3) dashes denote differentiation with respect to y , and

$$K = \omega/c. \quad (4)$$

For the region $y \geq 0$ (where M is a constant, M_0), equation (3) reduces to

$$P'' + k_y^2 P = 0, \quad k_y^2 = (K - k_x M_0)^2 - (k_x^2 + k_z^2). \quad (5, 6)$$

The general solution of equation (5) is

$$P = P_i \exp(ik_y y) + P_r \exp(-ik_y y). \quad (7)$$

If k_y is real and positive, the first term represents plane waves incident upon the boundary layer, and the second term represents reflected waves.

The problem of the reflection of plane waves from a plane surface is to find P_r when given P_i . In other words, one seeks the reflection coefficient

$$C_r = P_r/P_i. \quad (8)$$

The boundary layer acts much as a locally reacting solid boundary; one merely needs to know its acoustic admittance in order to calculate the reflection coefficient. One can define $Q(y)$ to be

$$Q(y) = P'/KP, \quad (9)$$

and denote the values of Q at the interface, $y = 0$, and at the wall, $y = -\delta$, by Q^* and Q_w , respectively. These quantities are proportional to the acoustic admittances of their respective planes. The acoustic admittance, χ , is by definition the ratio of the acoustic particle velocity normal to the plane to the acoustic pressure. For the pressure field described by equation (2), the particle velocity in the y -direction is

$$v = V(y) \exp[i(\omega t - k_x x - k_z z)]. \quad (10)$$

Therefore, the admittance of the plane, $y = 0$, is

$$\chi^* = V(0)/P(0). \quad (11)$$

The momentum equation shows that the velocity function, $V(y)$, is proportional to the derivative of the pressure function, $P'(y)$:

$$V(y) = iP'/\rho c(K - k_x M). \quad (12)$$

Combining equations (9), (11) and (12) provides the relation between Q^* and χ^* :

$$\rho c \chi^* = -iQ^*\{1 - (k_x/K)M_0\}. \quad (13)$$

Q_w and the admittance of the wall, χ_w , are related by a similar equation:

$$\rho c \chi_w = -iQ_w. \quad (14)$$

From equation (7) one can form the ratio of P' to P at $y = 0$, and find that

$$Q^* = ik_y(P_i - P_r)/K(P_i + P_r), \quad (15)$$

from which

$$C_r = (ik_y - KQ^*)/(ik_y + KQ^*). \quad (16)$$

Thus, the reflection coefficient is found when Q^* has been calculated.

This calculation can be accomplished by integration of equation (3) across the boundary layer from $y = -\delta$ to $y = 0$. However, the acoustic pressure itself is of secondary interest; the ratio of pressure gradient to pressure provides the acoustic admittance. Therefore, rather than attacking equation (3) directly, one can divide it by P and make the substitutions

$$P'/P = KQ, \quad P''/P = KQ' + K^2Q^2. \quad (17, 18)$$

This procedure transforms the second order equation into the Riccati equation,

$$KQ' + K^2Q^2 + 2Kk_x M'Q / (K - k_x M) + (K - k_x M)^2 - (k_x^2 + k_z^2) = 0. \quad (19)$$

It is now convenient to shift the origin to the wall and to make the independent variable dimensionless by means of the substitution,

$$s = (y + \delta)/\delta. \quad (20)$$

This transforms equation (19) into

$$\frac{dQ}{ds} + \frac{2k_x}{(K - k_x M)} \frac{dM}{ds} Q + K\delta \left[Q^2 + \left(1 - \frac{k_x}{K} M\right)^2 - \left(\frac{k_x}{K}\right)^2 - \left(\frac{k_z}{K}\right)^2 \right] = 0, \quad (21)$$

and integration should proceed from $s = 0$ to $s = 1$. A similar technique involving the Riccati equation has been applied to problems associated with wave propagation in nozzles and rocket engine instability [6].

The integration may be carried out only when the velocity profile, $M(s)$, has been specified and values of several parameters have been assigned. For the velocity profile one can choose

$$M = M_0 \quad \text{when } s \geq 1, \quad M = M_0 \sin(\pi s/2) \quad \text{when } 0 \leq s \leq 1, \quad (22, 23)$$

following Eversman [4] and Mungur and Plumlee [3]. The parameters are as follows: (1) the Mach number in the uniform flow, M_0 ; (2) the specific wall admittance, $\rho c \chi_w$, which provides, through equation (14), the value of Q at $s = 0$; (3) $K\delta$, which is a measure of the ratio of the boundary layer thickness to the wavelength the sound would have in a medium at rest; (4) the direction of incidence, which determines k_x/K and k_z/K . For the direction of incidence, only waves whose normals lie in the xy plane are considered; thus $k_z = 0$. If, with this assumption, one defines ϕ to be the angle between the normal to the incident plane wave and the y -axis, then

$$k_x/k_y = \tan \phi, \quad k_x/K = \sin \phi / (1 + M_0 \sin \phi), \quad (24, 25)$$

$$k_y/K = \cos \phi / (1 + M_0 \sin \phi). \quad (26)$$

Equations (25) and (26) are derived by combining equation (24) with equation (6).

In the limiting case of $K\delta = 0$, the governing differential equation (21) reduces to

$$\frac{dQ}{ds} + \frac{2k_x}{(K - k_x M)} \frac{dM}{ds} Q = 0, \quad (27)$$

which has the solution

$$Q / \{1 - (k_x/K)M\}^2 = Q_w. \quad (28)$$

Hence,

$$Q^* = i\chi_w \rho c / (1 + M_0 \sin \phi)^2, \quad C_r = \frac{\cos \phi (1 + M_0 \sin \phi) - \chi_w \rho c}{\cos \phi (1 + M_0 \sin \phi) + \chi_w \rho c}. \quad (29, 30)$$

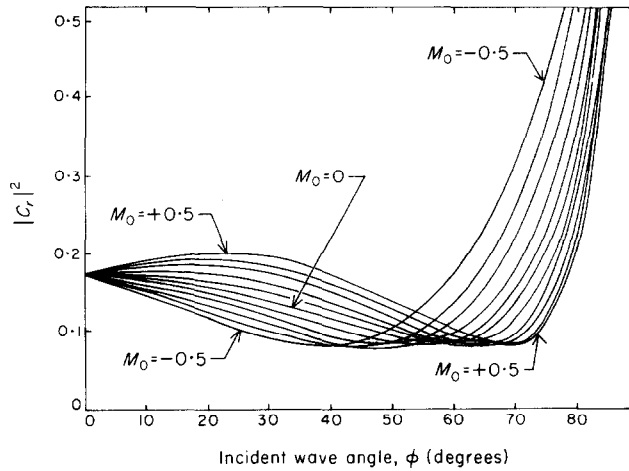


Figure 1. Fraction of energy reflected with slip flow, $K\delta = 0$. $\rho c\chi_w = 0.44943 - 0.28089i$, Mach number intervals of 0.1.

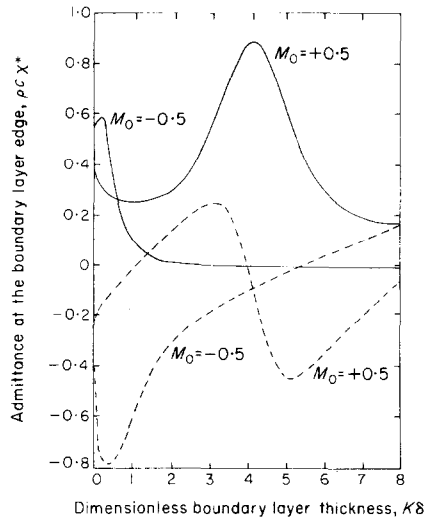
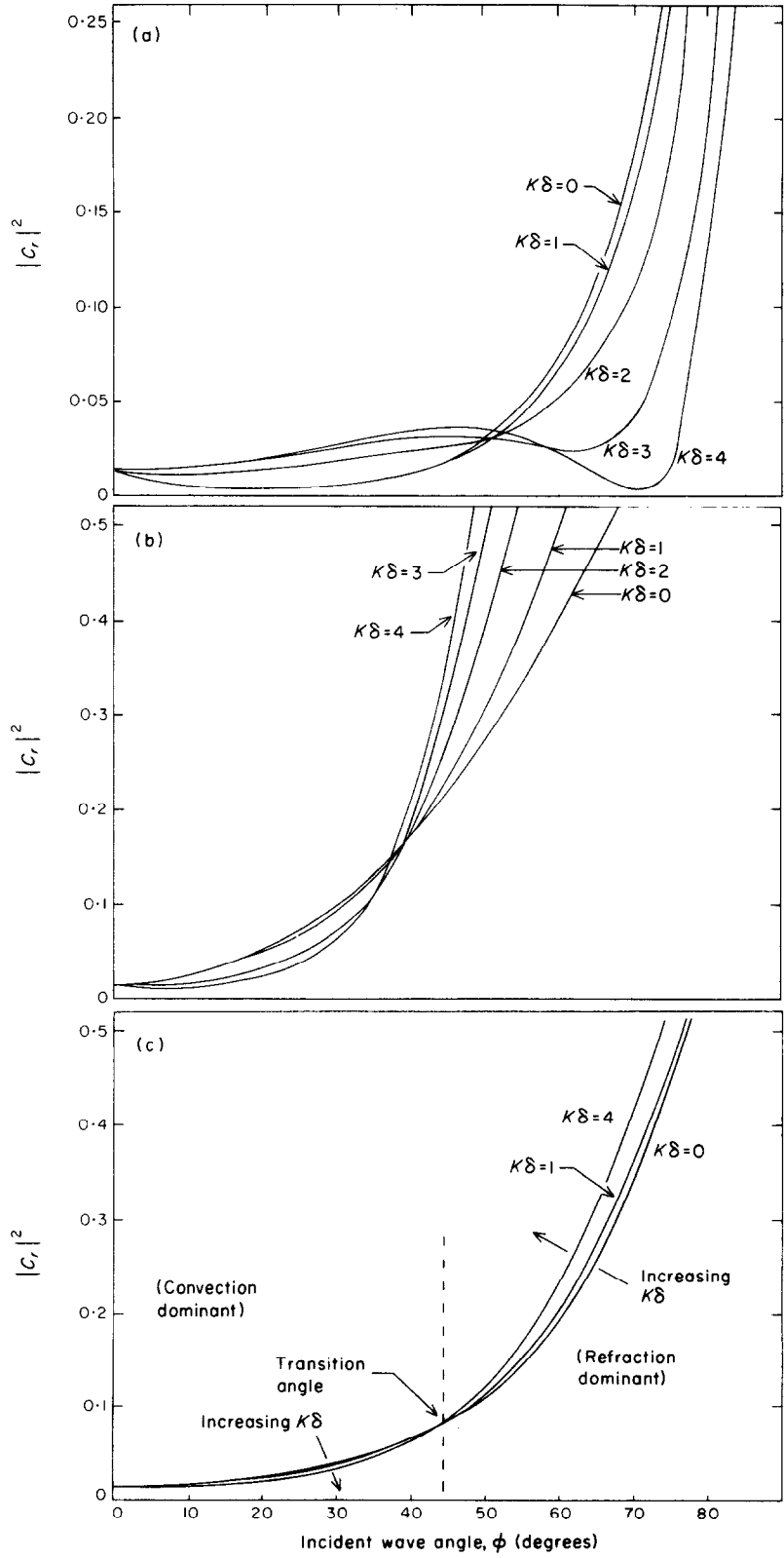


Figure 2. Admittance at the boundary layer edge when $\phi = 70^\circ$. $\rho c\chi_w = 0.44943 - 0.28089i$. —, Real part; ---, imaginary part.

Figure 1 shows the dependence of the reflection coefficient on the angle of incidence for several Mach numbers and one wall admittance. The sound waves travel in the direction of the positive x -axis. Sound propagation downstream or upstream is considered by assigning positive or negative values to the Mach number. These results for slip flow agree with those of Ingard [7].

For boundary layers of finite thickness, numerical integration of equation (21) is required. A Runge-Kutta procedure described by Hull, Enright and Jackson [8] produced the results shown in Figures 2–6. Figure 2 shows typical results for the boundary layer

Figure 3. Boundary layer effects on the fraction of energy reflected, for $\rho c\chi_w = 1.25 + 0.0i$. (a) $M_0 = +0.5$, downstream propagation; (b) $M_0 = -0.5$, upstream propagation; (c) $M_0 = -0.05$, near zero Mach number, upstream propagation.



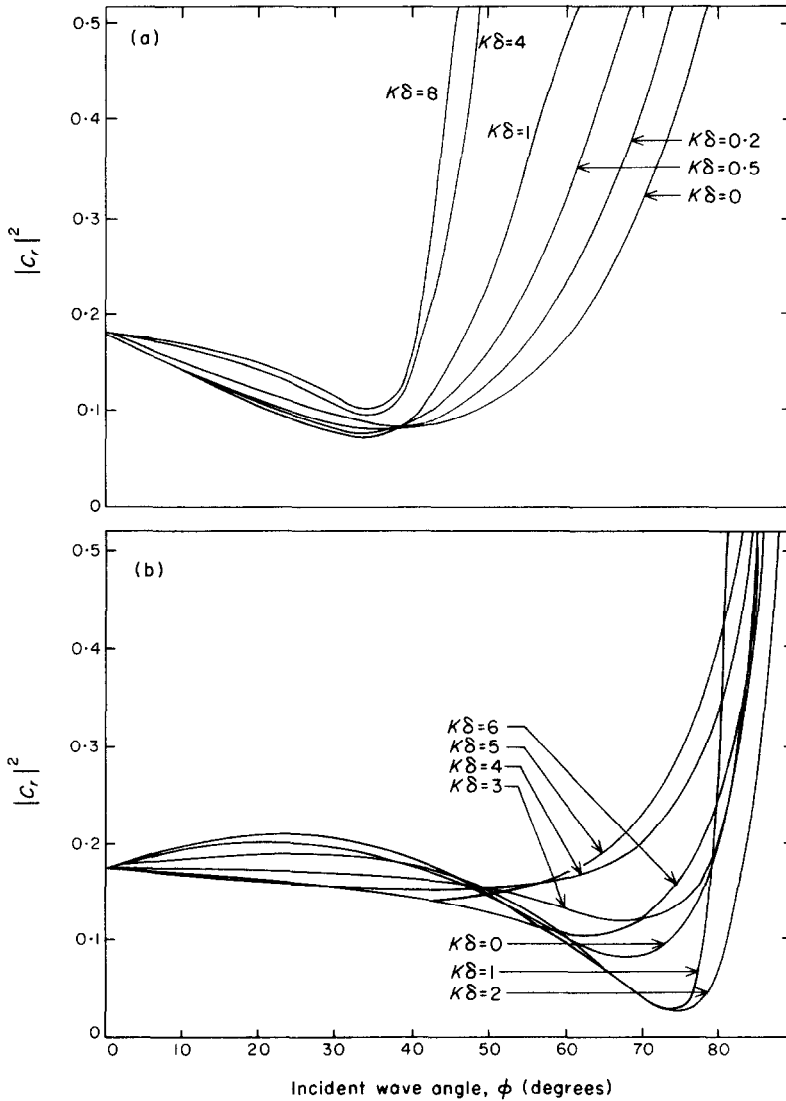


Figure 4. Boundary layer effects on the fraction of energy reflected, for $\rho c \chi_w = 0.44943 - 0.28089i$. (a) $M_0 = -0.5$, upstream propagation; (b) $M_0 = +0.5$, downstream propagation.

admittance, while Figures 3–6 show effects of the several parameters on the reflection coefficient.

Some of the effects can be explained qualitatively by considering the refraction of rays. When the sound propagates downstream, rays are refracted by the shear flow toward the wall. This makes the sound-absorbing material of the wall more effective and reduces the fraction of energy reflected. The effect is most pronounced at large angles of incidence. Thus, in Figure 3(a), for example, if ϕ is greater than 52° , the greater the boundary layer thickness, the more sound is absorbed. Figure 4(b) shows the same behavior, even though the wall admittance is quite different.

With upstream propagation the effect is the opposite. If the angle of incidence is great enough, rays are refracted away from the wall, and the sound-absorbing material is less

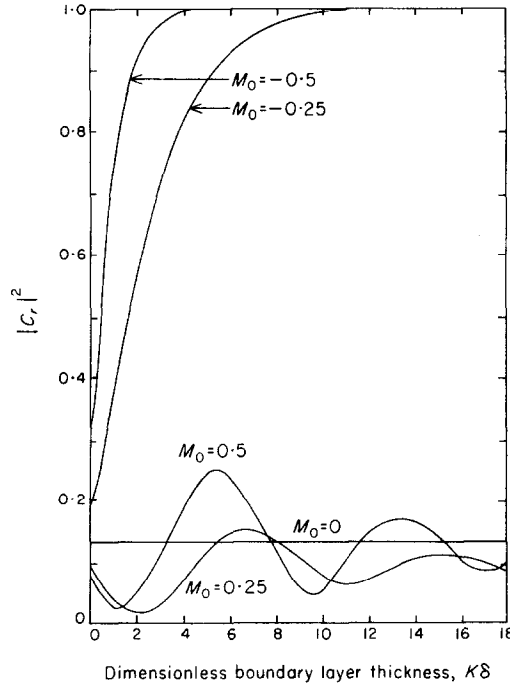


Figure 5. Variation of reflected energy with boundary layer thickness at $\phi = 70^\circ$. $\rho c \chi_w = 0.44943 - 0.28089i$.

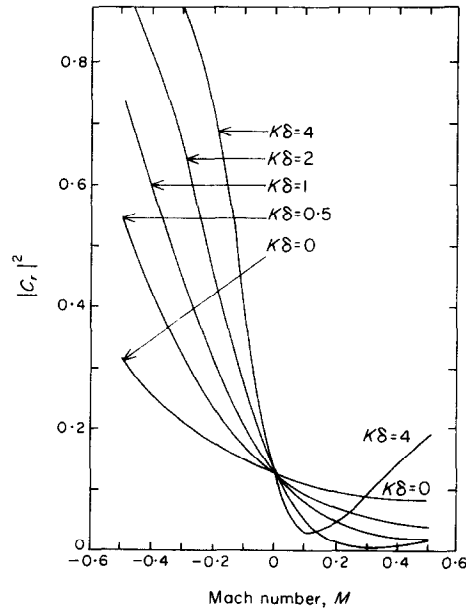


Figure 6. Dependence of reflection coefficient on Mach number at $\phi = 70^\circ$. $\rho c \chi_w = 0.44943 - 0.28089i$.

effective, as shown by Figures 3(b), 3(c) and 4(a). Indeed, for very large angles and thick boundary layers, refraction may prevent the sound from reaching the wall. All the energy is redirected into the steady flow region and the reflection coefficient is 1. This is shown by the upper two curves in Figure 5.

The effects of refraction are less if the boundary layer is thinner, the Mach number lower, or the angle of incidence smaller. Hence, as these parameters are reduced, the behavior is more like that of slip flow, as described by Ingard [7] and shown in Figure 1. One may call this regime "convection dominated", as contrasted with the "refraction dominated" behavior at larger values of ϕ , $K\delta$, and M_0 . In some cases a fairly clear transition from one regime to the other occurs; for example, at about 52° in Figure 3(a), 37° in Figure 3(b), and 44° in Figure 3(c).

Figure 5 illustrates the remarkable difference between upstream and downstream sound propagation when the angle of incidence is large and the boundary layer is thick.

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